

First Interim Report

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1 Topic and motivation

The topic of this project is *“What Drives Airbnb Occupancy and Pricing in European Markets? Evidence on Price Sensitivity, Booking Constraints, and Host Reputation.”* The research is conducted by Anna Kryvulia, Yustyna Hayevska, and Lidiia Sokha.

The rapid expansion of digital platforms such as Airbnb has significantly reshaped short-term rental markets across European countries. With increasing competition among hosts, understanding the factors that influence listing performance has become crucial. Hosts must make strategic decisions regarding pricing, booking rules, and service quality in order to attract guests and maximize occupancy, yet the effectiveness of these decisions is not always clear.

The motivation of this research is to empirically investigate how economic and behavioral factors determine demand in the Airbnb market across Europe. In particular, the study focuses on three key dimensions: price sensitivity, booking constraints, and host reputation. While price remains a fundamental determinant of demand, platform-specific features such as minimum stay requirements and instant booking options may introduce additional frictions that affect user choices. At the same time, reputation mechanisms, including review scores and Superhost status, serve as important signals of quality and trust in online marketplaces.

Understanding these relationships is important from both academic and practical perspectives. From an economic standpoint, the project contributes to the analysis of demand formation and price elasticity in platform-based markets. From a business perspective, the findings can help Airbnb hosts and property managers optimize their pricing strategies, adjust booking policies, and improve reputation signals in order to increase occupancy and revenue.

The main audience for this research includes Airbnb hosts, short-term rental agencies, and property managers operating across European markets. Additionally, the results may be useful for platform analysts and policymakers interested in the functioning and efficiency of digital accommodation markets.

2 The aim and the tasks

The main aim of this research is to analyze the factors that influence both the occupancy and the prices of Airbnb listings in Europe, and to estimate the impact of price sensitivity, booking constraints, host reputation, and listing characteristics using econometric models.

Tasks:

1. Identify the key variables that describe Airbnb listings in Europe, such as property type, number of rooms, neighborhood location, and host attributes like Superhost status.
2. Analyze how listing characteristics, location, and host-related factors affect Airbnb listing prices.
3. Analyze how pricing strategies and price elasticity affect the occupancy rate and overall demand for short-term rentals.
4. Investigate the impact of booking constraints, including minimum stay requirements, "Instant Book" settings, and cancellation policies, on listing performance.

5. Evaluate the role of reputation signals, specifically how the volume and quality of recent reviews influence listing prices and future booking trends.
6. Build econometric models to estimate the statistical significance and economic interpretation of the factors driving Airbnb prices and occupancy.

Based on these tasks, the study tests the following hypotheses:

- **Price Sensitivity:** Higher nightly prices are negatively associated with occupancy rates, suggesting that demand for Airbnb listings in Europe is price-sensitive.
- **Booking Friction:** Stricter booking constraints, such as longer minimum stay requirements and the absence of "Instant Book" functionality, significantly reduce listing occupancy.
- **Reputation and Social Proof:** Positive host reputation (Superhost status) and a high frequency of recent reviews have a stronger positive impact on occupancy than the overall historical rating.
- **Price Formation:** Listing characteristics, location advantages, and stronger host reputation are positively associated with higher Airbnb prices.

The results will allow us to better understand the dynamics of the European short-term rental market and provide practical insights for hosts seeking to improve their pricing, occupancy, and revenue strategies.

3 Literature review

Researches on Airbnb market:

1. Gibbs et al. - Pricing in the Sharing Economy: A Hedonic Pricing Model Applied to Airbnb Listings
This paper analyzes how listing characteristics, location, and host-related factors influence Airbnb prices and performance. It provides a strong foundation for understanding pricing and demand determinants.
Link: https://www.researchgate.net/publication/316056041_Pricing_in_the_sharing_economy_a_hedonic_pricing_model_applied_to_Airbnb_listings
2. Ert et al. - Trust and Reputation in the Sharing Economy: The Role of Personal Photos in Airbnb
This research highlights the importance of reputation signals such as reviews and host characteristics in shaping consumer decisions and demand.
Link: https://www.researchgate.net/publication/315410754_Trust_and_Reputation_in_the_Sharing_Economy_The_Role_of_Personal_Photos_on_Airbnb

Econometric methods (OLS and regression analysis):

1. Wooldridge - Introductory Econometrics: A Modern Approach A fundamental textbook that explains the use of OLS regression models, interpretation of coefficients, and causal inference in cross-sectional data.
Link: https://cbpbu.ac.in/userfiles/file/2020/STUDY_MAT/ECO/2.pdf

Price elasticity and log-models:

1. Varian - Intermediate Microeconomics: A Modern Approach Explains the concept of price elasticity of demand and its interpretation, which is central to analyzing price sensitivity in the Airbnb market.
Link: https://faculty.ksu.edu.sa/sites/default/files/microeco_varian.pdf

Fixed effects and applied econometrics:

1. Angrist and Pischke - Mostly Harmless Econometrics Provides practical guidance on applied econometric methods, including the use of fixed effects to control for unobserved heterogeneity.
Link: https://newblankets.org/worth_a_look/Mostly%20Harmless%20Econometrics.pdf

Fractional response models:

1. Papke and Wooldridge - Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates This paper introduces econometric methods for modeling dependent variables that are proportions bounded between 0 and 1.
Link: <https://onlinelibrary.wiley.com/doi/10.1002/%28SICI%291099-1255%28199611%2911%3A6%3C619%3A%3AAID-JAE418%3E3.0.CO%3B2-1>

4 Data description

To study the relationship between listing characteristics, pricing policies, reviews, and market outcomes on Airbnb, we use publicly available data provided by Inside Airbnb.

This dataset contains detailed information on Airbnb listings, daily calendar observations, and user reviews for selected cities.

The dataset is appropriate for this research because it combines structural characteristics of listings with time-related information.

The listings table includes variables such as property type, room type, number of bedrooms, capacity, neighborhood, host status, and review indicators. The calendar table contains daily information on availability and booking restrictions, which makes it possible to analyze occupancy patterns over time. The reviews table includes the dates of individual reviews and allows us to observe the flow of customer feedback over time.

At the same time, the data has an important limitation. The dataset does not directly report actual bookings. In the calendar table, unavailable dates may reflect either booked nights or dates blocked by hosts. Therefore, availability can only be used as a proxy for demand, not as a direct measure of it.

5 Methods

To analyze the factors that influence Airbnb listing prices and occupancy in European markets, this study applies four econometric models.

- 1) First, to examine the determinants of listing prices, we use a log-linear regression model:

$$\log(\text{Price}_i) = \beta_0 + \beta_1 X_i + u_i,$$

where X_i includes listing characteristics such as property type, room type, number of bedrooms, accommodation capacity, neighborhood, Superhost status, booking options, and review indicators. This model helps estimate how these factors are associated with price differences across listings.

- 2) Second, to analyze occupancy, we use a linear regression model. Since the dataset does not directly report bookings, occupancy is approximated by the share of unavailable days:

$$\text{Occupancy}_i = 1 - \frac{\text{availability_365}_i}{365}.$$

The corresponding model is:

$$\text{Occupancy}_i = \alpha_0 + \alpha_1 \ln(\text{Price}_i) + \alpha_2 Z_i + \varepsilon_i,$$

where Z_i includes booking constraints, reputation measures, and other listing controls. This model allows us to estimate which factors are associated with higher or lower occupancy.

- 3) Third, because the occupancy variable is a proportion bounded between 0 and 1, we also use a fractional response model:

$$E(\text{Occupancy}_i | X_i) = G(X_i \beta).$$

This model is more suitable for proportion-type dependent variables and serves as an alternative specification for occupancy.

In fact, this model is closely related to the logit model studied earlier in the course. The difference is that here the dependent variable is not binary; instead, it is a proportion bounded between 0 and 1. Therefore, the model is used to estimate the expected value of the dependent variable rather than the probability that it equals 1.

- 4) Finally, we apply a logistic regression model to estimate the probability that a listing belongs to the group of highly occupied listings:

$$P(Y_i = 1 | X_i) = \frac{\exp(X_i \beta)}{1 + \exp(X_i \beta)}.$$

This model helps identify which factors increase the probability of high occupancy.

Together, these methods allow us to study both main dimensions of Airbnb market performance in this project: listing prices and listing occupancy.

6 Schedule

1. The first interim for the topic. *Mar 28*
2. Data preparation and feature engineering.
Initial construction of price and occupancy models. *Apr 1*
3. Model refinement and preliminary evaluation (model quality and prediction on test dataset) *Apr 3*
4. Meeting to review results and plan final analysis *Apr 7*
5. Cross-check of price and occupancy models *Apr 12*
6. Comparison of model performance and interpretation of results *Apr 17*
7. Finalizing report and business interpretation of results *Apr 19*